

Problem Set 2: Detailed Answer Key & Derivations

CBE 30235: Introduction to Nuclear Engineering

Problem 1: C-14 Production Kinematics

Reaction: $^{14}\text{N} + n \longrightarrow ^{14}\text{C} + p + Q$

(a) Calculate Q-value

$$Q = [(M(^{14}\text{N}) + m_n) - (M(^{14}\text{C}) + M(^1\text{H}))] \times 931.5 \text{ MeV/u}$$

Substitute the masses:

$$\text{Reactants} = 14.003074 + 1.008665 = 15.011739 \text{ u}$$

$$\text{Products} = 14.003242 + 1.007825 = 15.011067 \text{ u}$$

$$\Delta m = 15.011739 - 15.011067 = 0.000672 \text{ u}$$

$$Q = 0.000672 \text{ u} \times 931.5 \text{ MeV/u} = \mathbf{0.626 \text{ MeV}}$$

(b) **Endothermic or Exothermic?** Since $Q > 0$, energy is released. The reaction is **Exothermic**.

(c) **Kinetic Energy Sharing** Since the incident neutron has negligible energy (thermal), the total kinetic energy shared by the products (^{14}C and p) is exactly equal to the Q-value: **0.626 MeV**. *Note: Conservation of momentum dictates that the lighter particle (proton) receives the vast majority of this energy ($E_p \approx \frac{14}{15}Q \approx 0.58 \text{ MeV}$), allowing it to travel far enough to potentially be detected.*

Problem 2: Solar Modulation of Cosmic Rays

(a) **Unshielded Production Rate** The production rate P is directly proportional to the incident flux ϕ .

$$\frac{P_{\text{new}}}{P_{\text{old}}} = \frac{\phi_{\text{interstellar}}}{\phi_{\text{solar}}} \approx 10$$

$$P_{\text{new}} = 10 \times 2.0 = \mathbf{20.0 \text{ atoms/(cm}^2 \cdot \text{s)}}$$

(b) **Solar Activity and the C-14 Clock** High solar activity corresponds to a stronger solar wind and a more turbulent heliospheric magnetic field.

- This **enhanced magnetic shielding** deflects more low-energy galactic cosmic rays (GCRs) away from Earth.
 - Result: **Lower production** of ^{14}C than average.
 - *Consequence:* Wood grown during high solar activity years will start with less ^{14}C and thus appear "older" than it really is unless a calibration curve is used.
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Problem 3: The "Moly Cow" (Transient Equilibrium)

(a) Decay Constants

$$\lambda_P = \frac{\ln 2}{66 \text{ hr}} = \mathbf{0.0105 \text{ hr}^{-1}}$$

$$\lambda_D = \frac{\ln 2}{6.0 \text{ hr}} = \mathbf{0.1155 \text{ hr}^{-1}}$$

(b) Transient Equilibrium Ratio Since $\lambda_P < \lambda_D$ (but $\lambda_P \neq 0$), this is Transient Equilibrium. The general solution for the daughter activity (assuming $N_D(0) = 0$) is:

$$A_D(t) = \frac{\lambda_D}{\lambda_D - \lambda_P} A_{P0} (e^{-\lambda_P t} - e^{-\lambda_D t})$$

At long times ($t \rightarrow \infty$), the term $e^{-\lambda_D t}$ becomes negligible compared to $e^{-\lambda_P t}$.

$$A_D(t) \approx \frac{\lambda_D}{\lambda_D - \lambda_P} A_{P0} e^{-\lambda_P t} = \frac{\lambda_D}{\lambda_D - \lambda_P} A_P(t)$$

The ratio becomes constant:

$$\frac{A_D}{A_P} = \frac{\lambda_D}{\lambda_D - \lambda_P} = \frac{0.1155}{0.1155 - 0.0105} = \frac{0.1155}{0.105} = \mathbf{1.10}$$

Note: The daughter activity is actually 10% higher than the parent activity!

(c) Time of Maximum Activity (t_{max}) To find the maximum, set $\frac{dA_D}{dt} = 0$. This yields the standard formula:

$$t_{max} = \frac{\ln(\lambda_D/\lambda_P)}{\lambda_D - \lambda_P}$$

$$t_{max} = \frac{\ln(0.1155/0.0105)}{0.1155 - 0.0105} = \frac{\ln(11)}{0.105} = \frac{2.3979}{0.105} \approx \mathbf{22.8 \text{ hours}}$$

Practical Implication: The hospital should milk the generator roughly once every 24 hours to maximize yield.

Problem 4: Spallation vs. Fission

(a) Comparison

- **Energy:** Fission is typically triggered by thermal neutrons ($\sim 0.025 \text{ eV}$). Spallation requires high-energy projectiles (protons) in the **GeV range** (10^9 eV).
- **Products:** Fission produces two large mass-asymmetric fragments and 2-3 neutrons. Spallation is a violent collision that shatters the nucleus, producing a "shower" (cascade) of dozens of neutrons, protons, pions, and muons.

(b) Choice of Target Material (High Z) At spallation sources (like SNS), we use heavy targets (Mercury, Tungsten) because neutron yield scales with atomic mass (A).

1. **Neutron Excess:** Heavy nuclei have a high N/Z ratio (roughly 1.5), providing a rich source of neutrons to be boiled off.

2. **Cross Section:** The geometric cross section for the incident proton is larger, increasing the probability of interaction.
 3. A single GeV proton on Mercury can release **20–30 neutrons**, whereas on Nitrogen it would yield far fewer.
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Problem 5: Atmospheric Attenuation

(a) **Survival Probability** Using the mean free path $\lambda = 90 \text{ g/cm}^2$ and total thickness $X = 1030 \text{ g/cm}^2$:

$$P(\text{no collision}) = e^{-X/\lambda} = e^{-1030/90} = e^{-11.44}$$

$$P \approx 1.07 \times 10^{-5} \quad (\approx 0.001\%)$$

(b) **Location of Production** Since the probability of a primary cosmic ray reaching sea level un-collided is essentially zero, virtually all high-energy protons interact in the upper atmosphere.

- The primary interaction creates a "shower" of secondary neutrons.
- These neutrons thermalize and capture on Nitrogen ($^{14}\text{N}(n, p)^{14}\text{C}$) locally.
- Therefore, the peak production of ^{14}C occurs at altitudes of 15–20 km (the stratosphere), where the cosmic ray flux is still high but the air density is sufficient to provide targets.